# Design of Enterprise IoT

## Tutorial 1

## Solution Scoping

# Smart Toll Collection (STC) System



Please read the use scenario description and complete the value proposition template

# Use Scenario Description (1/3)

To enhance Taiwan's transportation system and to improve traffic efficiency in all major roads, the government decided to develop an intelligent system that will be responsible for constructing a smart traffic solution. It is expected that this intelligent system will collect data from advanced sensors, and utilise communications and other technologies to provide real-time information that will increase transportation safety and efficiency.

# Use Scenario Description (2/3)

A major initiative in the context of the above project is to address highway toll collection issues. Previously, the manual toll collection stations were running in all the highways in Taiwan. These toll stations would process more than one million transactions daily. When passing through these toll stations, vehicles would have to reduce their speed to pay the toll with cash or toll tickets. During peak hours and holidays, the slow movement going to these toll stations usually become the source of traffic congestion.

# Use Scenario Description (3/3)

To solve this toll collection inefficiency, an IoT solution is required to be developed to address inefficient toll collection problem. The project, called Smart Toll Collection (STC) system, is considered one of the most important subprojects under the main intelligent system, as it uses end-to-end technology to provide real-time information on all vehicles travelling on the highways and enhances traffic efficiency.

**Please refer to the *Value Proposition Template* on Canvas and discuss the proposed value (and different aspects of needs and risks etc) for the Smart Toll Collection (STC) system solution.**

- *What does **'People'** refer to?*

- ***Substitute***: What people currently do instead?
  - What is the situation without the proposed new solution?
- ***Benefits***: What does the solution do?
  - What benefits it can bring to end users (e.g. customers, stakeholders)?
- ***Features***: How does the solution work?
- ***Experience***: What does it feel like to use the solution (system, product, etc.)?
- ***Fears***: What would be the potential risks and/or issues of the proposed solution?
- ***Needs***: What are the hidden needs and/or rational drivers of using this solution?

# Discussion: Sample Value Proposition

For the use case of Smart Toll Collection System

*What does **'People'** refer to?*

❑ *Consider two entities: the **Driver/User** and the **Government***

***Substitute***: What people currently do instead?

*"… the **manual toll collection stations** were running in all the highways in Taiwan. These toll stations would process **more than one million transactions daily**. When passing through these toll stations, vehicles would have to reduce their speed to **pay the toll with cash or toll tickets**. During peak hours and holidays, the slow movement going to these toll stations usually become **the source of traffic congestion**."*

- People have to reduce speed when they drive through the toll stations. (User)
- People have to prepare cash or the prepaid toll tickets. (User)
- People experience traffic congestion during peak hours and holidays. (User)
- Manual collection and processing of a large amount of transactions leads to inefficiency and being error-prone. (Government)
- Traffic congestion causes inefficient use of the highways. (Government)
- Management of human resources and toll stations is ongoing. (Government)

## ***Benefits***: What does the solution do?

*"To solve this toll collection inefficiency,* ***an IoT solution*** *is required to be developed to* ***address inefficient toll collection problem****. The project, called Smart Toll Collection (STC) system, …, as it uses … on all vehicles travelling on the highways and* ***enhances traffic efficiency****."*

- The STC system aims to speed up toll collection and solve the traffic congestion. (User and Government)
- The system is able to save the travel time. (User)
- The system can provide better payment experience. (User)
- The system can ensure the pay-as-you-use fairness. (User and Government)
- The system can help save the cost of building and maintaining the toll stations. It can also help reduce the human resource management cost. (Government)
- The system can reduce the waste of those toll tickets, further leading to environment protection. (User and Government)
- The system can provide real-time toll and highway usage data. (Government)
- The system can help reduce the accident rates. (User and Government)

## ***Features***: How does the solution work?

*"… this Intelligent system will **collect data from advanced sensors, utilises communications and other technologies to provide real-time information** that will increase transportation safety and efficiency."*

*"The project, called Smart Toll Collection (STC) system, …, as it **uses end-to-end technology to provide real-time information on all vehicles travelling on the highways** and enhances traffic efficiency."*

- The STC system automatically detects the travelling vehicles passing a toll point and records the relevant information about the vehicles. (Government)
- The system computes the appropriate toll fee to charge on a travelling vehicle, and allows multiple ways to collect the payments. (User and Government)
- The system automatically notifies the travelling vehicles when passing the toll point and the charge of a toll fee. (User)
- The system collects data capturing real-time information about travelling vehicles and traffic status at toll points and analyses the data to help improve transportation safety and efficiency. (Government)

## ***Experience***: What does is feel like to use the solution?

- People will feel more convenient, safer, and fair when using the highway and paying the fee. (User)
- It will be more efficient, more convenient and safer for the government to collect toll fees and manage the traffic of the highway. (Government)

*Emotional drivers*

- People feel less bothered because they don't have to prepare cash or toll tickets. (User)
- People feel less annoyed because they don't have to reduce speed every certain distance. (User)
- People feel satisfied because they can travel freely and faster. (User)
- People feel fair because everyone is charged equally. (User)

## ***Fears***: What would be the potential risks of the proposed solution?

- There could be malfunction of the system which leads to the loss of the toll fee. (Government)
- There might be a slow adoption of the new system in the beginning because people need to adapt to it and cars have to be installed with the device. (User)
- People who are not tech-oriented, such as older people, may encounter problems when using new device and system. (User)
- It can lead to labour protest because the STC system does not require human operations at toll stations. All toll collectors will likely be laid off or moved to different positions. (Government)

## ***Needs***: What are the hidden needs & rational drivers of using this solution?

*(Hidden) needs*

- Government should check necessary improvement or upgrading for the existing infrastructure. (Government)
- Government should integrate resources to solve problems like system design, user education, societal impacts such as job loss, etc. (Government)
- Drivers should expect behaviour changes and learn how to use new technology. (User)

*Rational drivers*

- Both people and the government want more convenient and efficient toll collection and better traffic condition when travelling on highway. (User and Government)
- Behind the system is a large amount of collected data which can be used to generate valuable insight further for traffic analysis and other transportation policy development. (Government)
- Government can receive the deserved amount of the toll fee. (Government)

# Recap & Reflect

Using value proposition in requirement analysis ensures that requirements are practical, relevant, and well aligned with what stakeholders need.

- **People** (who will use the system)
  - Key stakeholders
- **Benefits** (what users gain from the system)
  - Defines why the system matters to users
- **Features** (what the system provides to users)
  - Translates user benefits into tangible system capabilities
- **Experience** (how users interact with the system)
  - Focuses on usability, accessibility, interaction design, and ease of adoption
- **Needs** (what must be addressed for employing the system)
  - Certain prerequisites (e.g. infrastructure readiness, operational adjustments, public acceptance) and rational drivers focusing on logical justification for adoption
- **Fears** (what are potential risks in employing the system)
- **Substitute** (what do users do without employing the system)
  - Scope: Describe the current state without the system, focusing on existing situation rather than comparing with competing solutions.

# Questions?